K L UNIVERSITY

FRESHMAN ENGINEERING DEPARTMENT

A Project Based Lab Report

On

LOST ITEM LOGGER SYSTEM

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CERTIFICATE

This is to certify that the project based laboratory report entitled "LOST ITEM LOGGER SYSTEM" submitted by **ASHUTOSH JHA , AARISHI DEY , KARRA NIKHIL, VUTU SRI LAKSHMAN bearing Regd. No. 2500030826, 2500030907, 2500030806, 2500031250 to the Department COMPUTER SCIENCE ENGINEERING , KL University** in partial fulfillment of the requirements for the completion of a project based Laboratory in **"PYTHON PROGRAMMING (25SC2007)"** course in I B Tech II Semester, is a Bonafide record of the work carried out by them under my supervision during the academic year 2025 – 2026.

PROJECT SUPERVISOR

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ACKNOWLEDGEMENTS

It is great pleasure for me to express my gratitude to our honorable President. **Sri. Koneru Satyanarayana**, for giving the opportunity and platform with facilities in accomplishing the project-based laboratory report.

We express sincere gratitude to our Director **Dr. A. Jagdeesh** for his administration towards our academic growth. We express sincere gratitude to our Principal **Dr. V. Krishna Reddy** for his administration towards our academic growth.

We express sincere gratitude to our HOD-BES-1 **Dr. D. Haritha** for her leadership and constant motivation provided for the successful completion of our academic semester. We record it as my privilege to deeply thank you for providing us with the efficient faculty and facilities to make our ideas into reality.

We express our sincere thanks to our project supervisor **Dr E Sreedevi** for her novel association of ideas, encouragement, appreciation, and intellectual zeal which motivated us to venture this project successfully.
Finally, We are pleased to acknowledge our indebtedness to all those who devoted themselves directly or indirectly to making this project a success.

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ABSTRACT

In a dynamic college environment, students and staff frequently face the problem of misplaced belongings such as ID cards, books, electronic devices, and personal accessories. The absence of a structured and centralized system for reporting and tracking these items often results in confusion, delayed recovery, or permanent loss.

To address this issue, the **Lost Item Logger System for College Campus (LIL-CAMP)** has been developed as an efficient and user-friendly solution using Python.
This project introduces a digital platform that allows users to systematically record both lost and found items with essential details including item name, description, location, date, category, and contact information.

Unlike traditional manual methods, the system ensures organized data storage using CSV and JSON formats, making retrieval and management easier and more reliable.

A key feature of this system is its intelligent matching mechanism, which compares lost and found entries based on similarities in item description, category, and location. This significantly increases the probability of recovering lost items. Additionally, the system supports advanced search and filtering options, enabling users to quickly locate relevant records.

The project also integrates data analysis and visualization tools such as Pandas, NumPy, and Matplotlib to generate meaningful insights.

These include identifying commonly lost items, high-risk locations, and overall recovery performance, which can help improve campus management strategies.

Beyond solving a real-world problem, this project demonstrates the practical application of core Python concepts such as object-oriented programming, file handling, exception handling, and modular design. Overall, LIL-CAMP serves as a smart, scalable, and impactful solution that enhances efficiency, transparency, and communication within the campus community.

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INTRODUCTION

1. Background and Problem Statement

In modern educational institutions, especially large college campuses, students and staff carry a wide variety of personal and academic belongings such as identity cards, mobile phones, laptops, books, wallets, keys, USB drives, and other accessories. Due to the dynamic nature of campus life, where individuals frequently move between classrooms, laboratories, libraries, cafeterias, and hostels, the chances of misplacing or losing items are significantly high. The problem of lost and found items is not new, but it remains largely unaddressed in an organized and systematic manner. Most institutions still rely on traditional and informal methods such as:

- Asking peers or classmates
- Checking notice boards
- Contacting faculty or administrative offices
- Posting messages in social media groups

These methods are inefficient, time-consuming, and unreliable. There is no guarantee that the information will reach the right person. Moreover, even when items are found, they often remain unclaimed due to lack of proper identification or communication channels. Another major issue is the absence of a **centralized database**. Without a proper system to store and retrieve information, records of lost and found items are scattered and unorganized. This reduces the probability of recovering lost belongings and leads to frustration among users. To overcome these limitations, there is a need for a **digital, structured, and intelligent system** that can efficiently manage lost and found items within a campus environment.

2. Introduction to the Proposed System

The **Lost Item Logger System for College Campus (LIL-CAMP)** is designed as a comprehensive solution to address the challenges associated with lost and found item management. It is a Python-based application that provides a centralized platform where users can:

- Report lost items
- Submit found items Search for items
- View records in an organized format

The system ensures that all data is stored digitally, making it easy to access, update, and manage. Unlike manual methods, this system improves efficiency by reducing

human effort and minimizing errors.
One of the key features of the system is its **matching mechanism**, which compares lost and found item entries based on attributes such as:

- Item name
- Description
- Category
- Location
- Date

This feature significantly enhances the chances of recovering lost items by automatically identifying possible matches.

3. Role of Python in the Project

Python is chosen as the primary programming language for this project due to its simplicity, flexibility, and powerful ecosystem of libraries. It is widely used in software development, data analysis, and automation.
The advantages of using Python include:

- Easy-to-read and understandable syntax
- Rapid development and prototyping
- Strong support for libraries and frameworks
- Platform independence (runs on Windows, Linux, Mac) The project makes use of several core Python concepts such as:
- **Object-Oriented Programming (OOP)** for structuring the system into classes and objects
- **File Handling** for storing and retrieving data
- **Exception Handling** for managing errors and ensuring smooth execution
- **Modular Programming** for dividing the system into manageable components

4. Use of NumPy, Pandas, and Matplotlib

To enhance the functionality of the system, advanced Python libraries are integrated:

NumPy (Numerical Python)
NumPy is used for efficient numerical computations and data processing. It helps in:

- Performing statistical analysis
- Calculating frequencies of lost items
- Handling large datasets efficiently **Pandas**

Pandas is a powerful library for data manipulation and analysis. In this project, it is used to:

- Organize data into structured formats (DataFrames)
- Filter and sort record
- Perform data cleaning and transformation

- Analyze trends in lost and found items

Matplotlib

Matplotlib is used for data visualization. It allows the system to generate graphical representations such as:

- Bar charts (most lost items)
- Pie charts (category distribution)
- Line graphs (loss trends over time)

These visualizations help in understanding patterns and making informed decisions.

5. System Features and Functionalities

The Lost Item Logger System includes several important features that make it effective and user-friendly:

1. Centralized Data Management

All records of lost and found items are stored in a single system, ensuring easy access and organization.

2. User Input Interface

Users can easily enter details such as:

- Item name
- Description
- Location
- Date
- Contact information

3. Search and Filter Functionality

The system allows users to search items using keywords, making it quick to locate relevant records.

4. Matching Mechanism

Automatically compares lost and found items to identify potential matches.

5. Data Storage

Data is stored using:

- JSON files CSV files

  This ensures persistence and easy retrieval.

6. Data Analysis

The system analyzes stored data to identify:

- Frequently lost items
- High-risk locations
- Recovery rates

Graphs and charts provide a visual understanding of data trends.

6. Significance of the Project

The significance of this project lies in its ability to solve a common real-world problem using technology. It provides several benefits:

- **Improves Efficiency:** Reduces time required to find lost items
- **Enhances Communication:** Connects people who lost and found items
- **Increases Recovery Rate:** Smart matching improves chances of retrieval
- **Maintains Records:** Keeps a digital log of all entries
- **Supports Decision Making:** Data analysis helps identify patterns

7. Real-World Applications

Although the system is designed for a college campus, it can be extended to various other environments:

- Schools and universities
- Offices and corporate organizations
- Airports and railway stations
- Shopping malls
- Public libraries

With further development, it can be converted into:

- A web application
- A mobile application
- A cloud-based system

8. Learning Outcomes and Technical Skills

This project helps in developing several technical skills:

- Understanding Python programming concepts
- Working with real-world data
- Implementing file handling techniques

9. Future Scope

The project has great potential for future enhancements, such as:

- Adding a **Graphical User Interface (GUI)**
- Developing a **web-based platform**
- Integrating **database systems like MySQL**
- Adding **user authentication and security features**
- Implementing **AI-based matching algorithms**
- Providing **real-time notifications**

Conclusion of Introduction

In conclusion, the Lost Item Logger System (LIL-CAMP) is a practical and efficient solution designed to address the common issue of lost and found items in a college campus. By leveraging Python and its powerful libraries, the system provides a structured, reliable, and intelligent platform for managing item records.

The integration of data analysis and visualization further enhances its usefulness by providing insights into patterns and trends. Overall, the project not only solves a real-world problem but also demonstrates the effective application of programming concepts in developing meaningful software solutions.

AIM

The aim of this project is to design and develop a **Lost Item Logger System for College Campus (LIL-CAMP)** using Python, which provides a centralized and efficient platform to **record, track, search, and match lost and found items**.

This system aims to:

- Reduce the time and effort required to find lost items
- Improve communication between students and staff
- Increase the chances of item recovery through smart matching
- Maintain a proper digital record of all lost and found items
- Apply and demonstrate key Python programming concepts in a real-world application

Overall, the goal is to create a simple, reliable, and user-friendly system that solves the common problem of lost items in a college campus.

ADVANTAGES

- **Centralized System**
  All lost and found items are stored in one place, making tracking easy and organized.
- **Saves Time and Effort**
  Users can quickly search for items instead of manually asking or checking different places.
- **Improves Recovery Rate**
  The matching feature increases the chances of returning items to their owners.
- **Easy to Use**
  Simple interface and menu-driven system make it user-friendly for students and staff.
- **Data Analysis & Reports**
  Helps identify frequently lost items and high-risk locations using graphs and statistics.
- **Digital Record Maintenance**
  Keeps a proper record of all entries, avoiding loss of information.
- **Learning Experience**
  Covers important Python concepts like OOP, file handling, and data analysis.

DISADVANTAGES

- **Requires Basic Technical Knowledge**
  Users need some understanding of the system to operate it.
- **Dependence on User Input**
  Incorrect or incomplete data can affect matching accuracy.
- **Limited Accessibility (if offline)**
  If not deployed online, access may be restricted to a single system.
- **Data Security Concerns**
  Personal contact information must be handled carefully to avoid misuse.
- **Matching Not Always Accurate**
  Similar items may not always be correctly matched.

Initial Setup Time
Setting up the system and database requires time and effort..

FUTURE IMPLEMENTATION

The system is implemented using **Python programming concepts** step-by-step:

1. **Basic Input System**
   - User enters item details like name, location, date, etc.
2. **Menu-Driven Program** ▯ Options to:
   - Add lost item
   - Add found item
   - Search item
3. **Data Storage**
   - Data stored using:
   - Lists and Dictionaries
   - CSV/JSON files
4. **Object-Oriented Programming**
   - Created Item class
   - Used encapsulation and inheritance
5. **File Handling**
   - Save and load data from files
6. **Error Handling**
   - Handles invalid input, missing fields, etc.
7. **Data Analysis**
   - Used Pandas to analyze lost/found data
8. **Visualization**
   - Used Matplotlib to show graphs

SYSTEM REQUIREMENTS

➢ SOFTWARE REQUIREMENTS:

The software requirements needed to develop and run this project are:

- **Programming Language:** Python
- **Operating System:** Windows 10 or above (can also run on Linux/Mac)
- **Libraries/Modules Used:**
  - Pandas (for data analysis)
  - NumPy (for numerical operations)
  - Matplotlib (for data visualization)
- **Development Tools:**
  - VS Code / PyCharm / IDLE (any Python IDE)

➢ HARDWARE REQUIREMENTS:

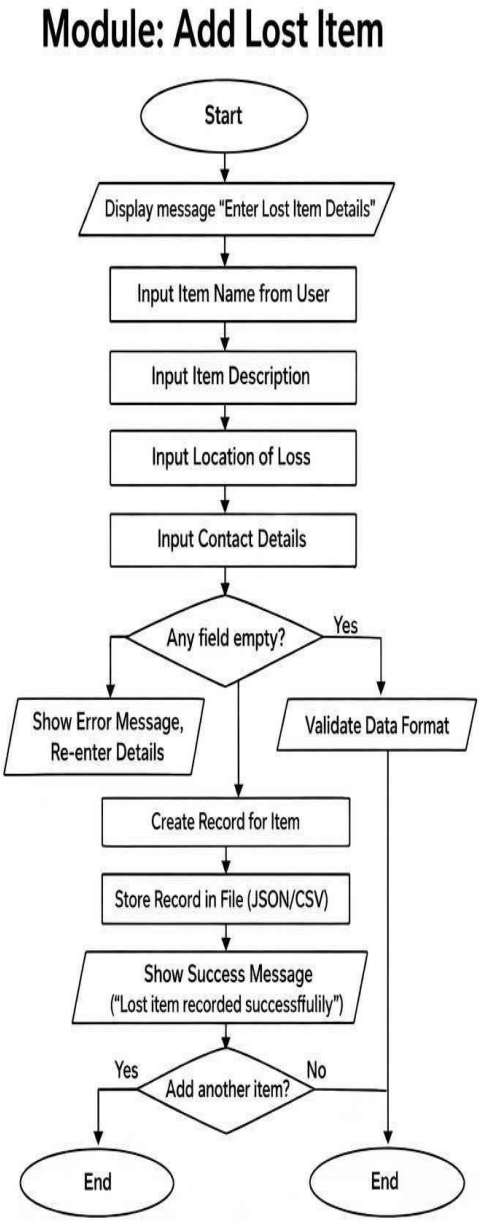
Processor: Intel i3 or above

RAM: Minimum 2 GB (4 GB recommended for smooth performance)

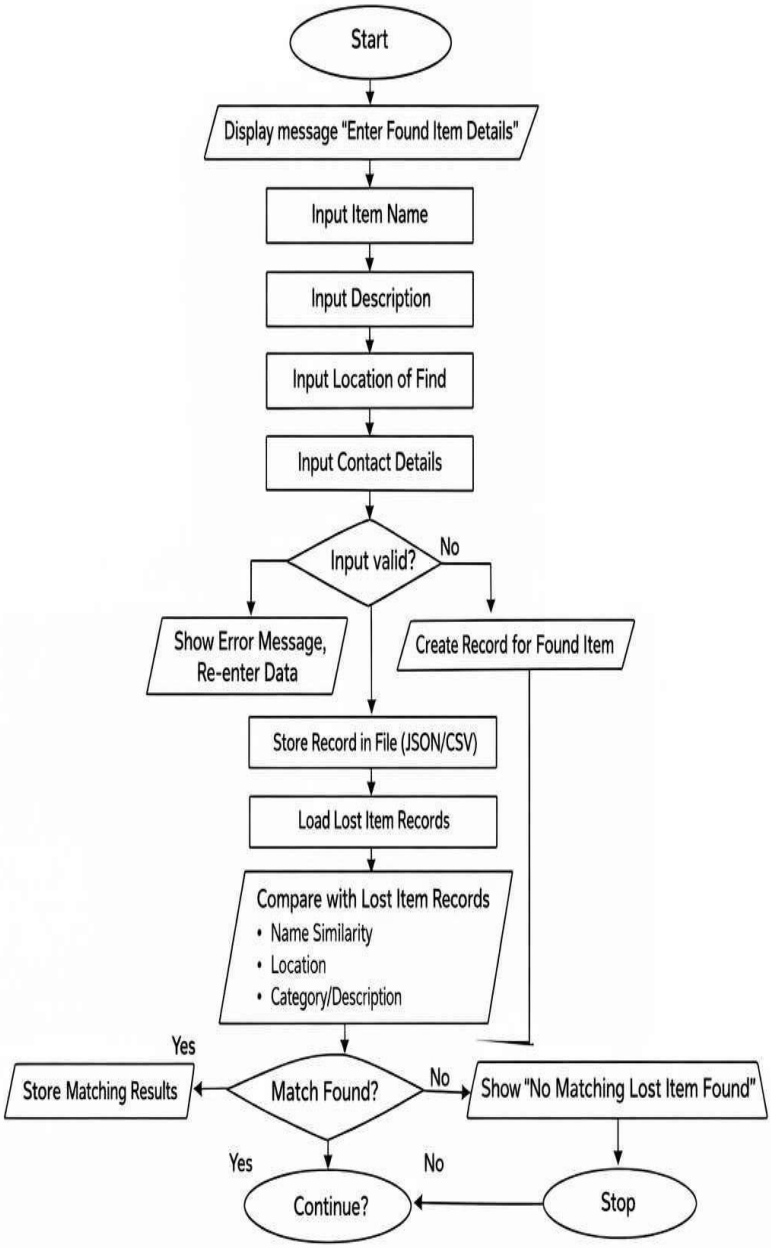
Input Devices: Keyboard, Mouse

Output Device: Monitor

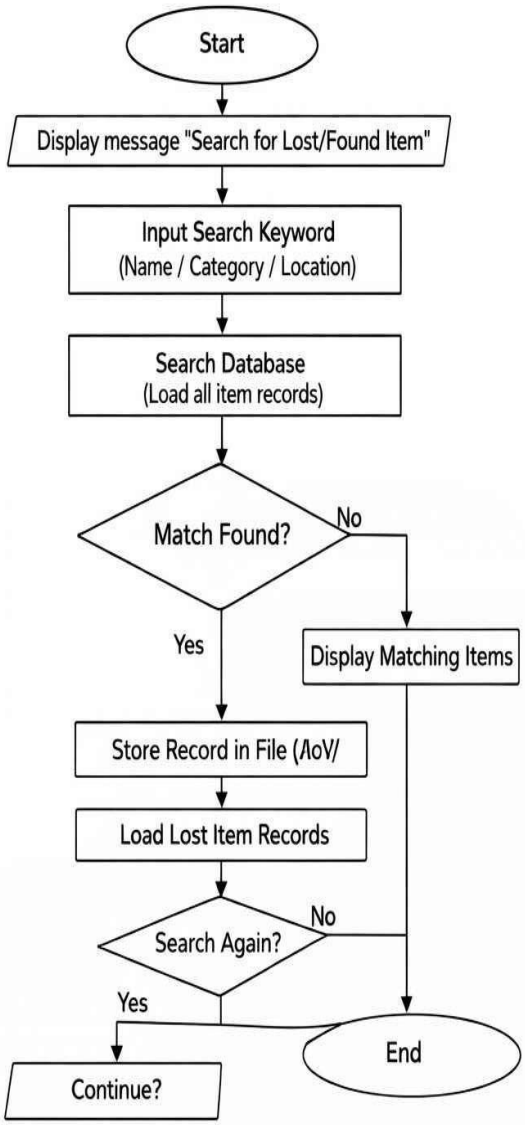
DATA FLOW DIAGRAM

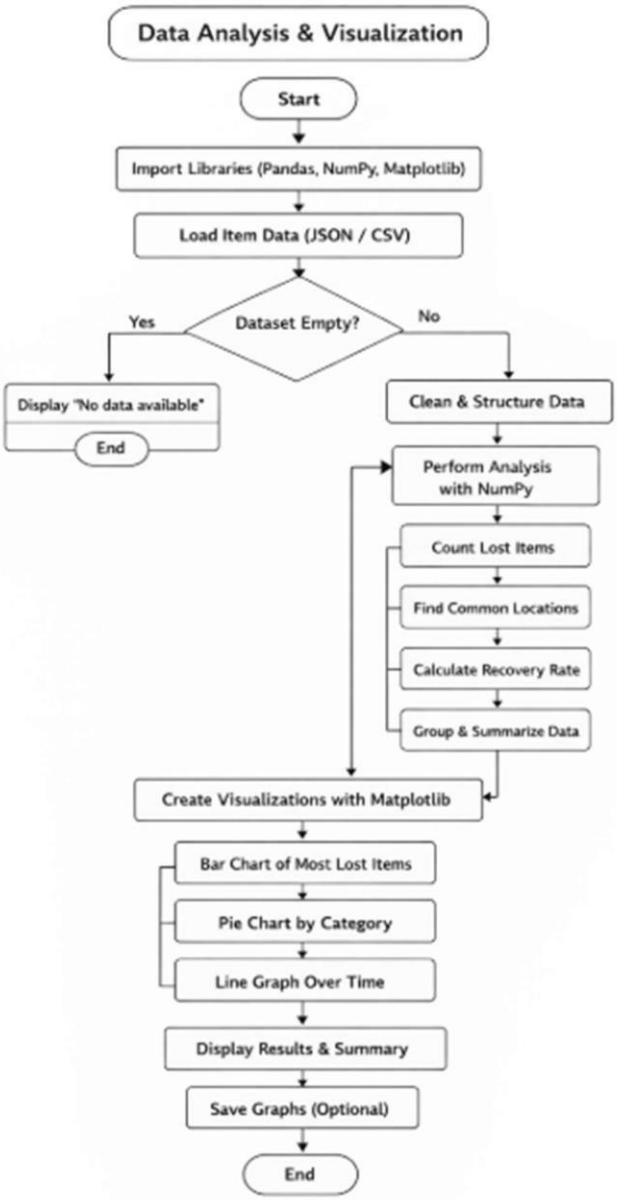


Module: Add Found Item



Module: Search Item




Data Analysis & Visualization
Start
Import Libraries (Pandas, NumPy, Matplotlib)
Load Item Data (JSON / CSV)
Dataset Empty?
Yes
No
Display "No data available"
End
Clean & Structure Data
Perform Analysis with NumPy
Count Lost Items
Find Common Locations
Calculate Recovery Rate
Group & Summarize Data
Create Visualizations with Matplotlib
Bar Chart of Most Lost Items
Pie Chart by Category
Line Graph Over Time
Display Results & Summary
Save Graphs (Optional)
End

ALGORITHM

1. MODULE: ADD LOST ITEM

Algorithm

1. Start
2. Display message "Enter Lost Item Details"
3. Input item name from user
4. Input item description
5. Input location where item was lost
6. Input date of loss
7. Input user contact details
8. Check if any field is empty
9. If empty → display error message and ask to re-enter details
10. Else validate data format (date, text, etc.)
11. Create a record (dictionary/object) for the item
12. Open file (JSON/CSV) in write/append mode
13. Store the item details in the file
14. Close the file
15. Display confirmation message "Lost item recorded successfully"
16. Ask user if they want to add another item
17. If yes → repeat from Step 3
18. Else → proceed to end
19. Stop

2. MODULE: ADD FOUND ITEM

Algorithm

1. Start
2. Display message "Enter Found Item Details"
3. Input item name
4. Input description
5. Input location where item was found
6. Input date of finding
7. Input contact details
8. Check for missing or invalid input
9. If invalid → display error and re-enter data

10. Else proceed further
11. Create a record for the found item
12. Open file and store the data
13. Load all lost item records from file
14. Compare found item details with lost item records based on:
- Name similarity
- Location
- Category/description
15. If match found → store matching results
16. Display matched items to the user
17. If no match found → display "No matching lost item found"
18. Ask user if they want to continue
19. If yes → repeat process
20. Else → Stop

3. MODULE: SEARCH ITEM

Algorithm

1. Start
2. Display message "Search for Lost/Found Item"
3. Input search keyword (name/category/location)
4. Convert keyword to standard format (lowercase, trimmed)
5. Load all item records from file
6. Initialize an empty list for results 7. Traverse each record in the database
8. Compare keyword with:
    - Item name
    - Description
    - Location
9. If match found → add item to result list
10. Continue searching until all records are checked
11. If result list is not empty: ▯ Display all matching items
12. Else:

▯ Display "No items found"

13. Ask user if they want to search again
14. If yes → repeat from Step 3
15. Else → Stop

4. MODULE: DATA ANALYSIS & VISUALIZATION

Algorithm

1. Start
2. Import required libraries (Pandas, NumPy, Matplotlib)
3. Load item data from JSON/CSV file using Pandas
4. Check if dataset is empty
5. If empty → display "No data available" and stop
6. Clean the dataset (remove duplicates, handle missing values)
7. Convert data into structured format (DataFrame)
8. Use NumPy to perform analysis:
    - Count frequency of lost items
    - Identify most common locations
    - Calculate recovery rate
9. Group data based on categories (item type/location)
10. Generate statistical summaries 11. Use Matplotlib to create graphs:

- Bar chart for most lost items
- Pie chart for category distribution
- Line graph for trends over time

12. Label graphs properly (title, axes, legend)
13. Display graphs to the user
14. Save graphs (optional)
15. Display analysis summary
16. Stop

IMPLEMENTATION

```
import json

FILE = "lost_items.json"

# Load
data def
load_items():
try:
with open(FILE, "r") as f:
return json.load(f)
except:
return []

# Save data
def
save_items(items):
with open(FILE, "w") as f:
json.dump(items, f)

# Backup
def
backup():
with open("backup.json", "w") as f:
json.dump(load_items(), f)


# Main Program
lost_items
load_items()

while True:
print("\n1. Report Lost Item")
print("2. View Lost Items")
print("3. Exit")

choice = int(input("Enter your choice:
"))

if choice == 1:
```

```
item = {
"item_name": input("Enter Item Name: "),
"color": input("Enter Item Color: ")

"location": input("Enter Place Lost: "), "date": input("Enter Date (DD-MM-YYYY): "),
"contact": input("Enter Student Name: ")
}

lost_items.append(item)
save_items(lost_items)

print("\nLost item has been successfully recorded.")
print("Thank you for helping maintain campus records.")

elif choice == 2:
if len(lost_items) == 0:
print("\nNo lost items reported yet.")
else:
print("\n--------- Lost Item Details \n")

for item in lost_items:
print("Item Name :", item["item_name"])
print("Color :", item["color"])
print("Lost At :", item["location"]) print("Date
:", item["date"])
print("Reported By :", item["contact"])
print("\n ")
elif choice == 3:
backup()
print("\nThank you for using the system.") break

else:
print("\nInvalid choice. Please try again.")
```

INTEGRATION AND SYSTEM TESTING OUTPUTS

Screen Shots:

```
1. Report Lost Item
2. View Lost Items
3. Exit
Enter your choice: 1
Enter Item Name: Python Workbook
Enter Item Color: Grey
Enter Place Lost: C108
Enter Date (DD-MM-YYYY): 2-02-2026
Enter Student Name: Nispesita

Lost item has been successfully recorded.
Thank you for helping maintain campus records.

1. Report Lost Item
2. View Lost Items
3. Exit
Enter your choice: 1
Enter Item Name: Bottle
Enter Item Color: Pink
Enter Place Lost: F 107
Enter Date (DD-MM-YYYY): 4-03-2026
Enter Student Name: Sonam

Lost item has been successfully recorded.
Thank you for helping maintain campus records.

1. Report Lost Item
2. View Lost Items
3. Exit
Enter your choice: 1
Enter Item Name: Headphones
```

```
Enter Item Color: Blue
Enter Place Lost: Ground
Enter Date (DD-MM-YYYY): 7-03-2026
Enter Student Name: Ashutosh

Lost item has been successfully recorded.
Thank you for helping maintain campus records.

1. Report Lost Item
2. View Lost Items
3. Exit
Enter your choice: 1
Enter Item Name: Room keys
Enter Item Color: Grey
Enter Place Lost: Rnd block
Enter Date (DD-MM-YYYY): 10-03-2026
Enter Student Name: Shubham

Lost item has been successfully recorded.
Thank you for helping maintain campus records.

1. Report Lost Item
2. View Lost Items
3. Exit
Enter your choice: 2

--------- Lost Item Details ---------

Item Name   : Python Workbook
Color       : Grey
Lost At     : C108
Date        : 2-02-2026
Reported By : Nispesita
```

```
-----------------------------------
Item Name   : Bottle
Color       : Pink
Lost At     : F 107
Date        : 4-03-2026
Reported By : Sona,

-----------------------------------
Item Name   : Headphones
Color       : Blue
Lost At     : Ground
Date        : 7-03-2026
Reported By : Ashutosh

-----------------------------------
Item Name   : Room keys
Color       : Grey
Lost At     : Rnd block
Date        : 10-03-2026
Reported By : Shubham

-----------------------------------

1. Report Lost Item
2. View Lost Items
3. Exit
Enter your choice: 3

Thank you for using the system.

Process finished with exit code 0
```

CONCLUSION

The **Lost Item Logger System for College Campus (LIL-CAMP)** is a practical and efficient solution to the common problem of lost and misplaced items in a college environment. The system successfully provides a centralized platform where users can easily report lost items, submit found items, and search for them in an organized manner.

By using features like data storage, searching, and matching of items, the system improves the chances of recovering lost belongings and reduces the time and effort required in manual searching. The inclusion of data analysis and visualization also helps in understanding patterns such as frequently lost items and high-risk locations.

This project not only solves a real-world problem but also demonstrates the effective use of Python programming concepts such as object-oriented programming, file handling, exception handling, and modular design.

Overall, the system is simple, user-friendly, and reliable, and it can be further improved by adding features like a web interface or mobile application for better accessibility.